



A survey on privacy risks and protection in large language models

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Abstract

Although Large Language Models (LLMs) have become increasingly integral to diverse applications, their capabilities raise significant privacy concerns. This survey offers a comprehensive overview of privacy risks associated with LLMs and examines current solutions to mitigate these challenges. First, we analyze privacy leakage and attacks in LLMs, focusing on how these models unintentionally expose sensitive information through techniques such as model inversion, training data extraction, and membership inference. We investigate the mechanisms of privacy leakage, including the unauthorized extraction of training data and the potential exploitation of these vulnerabilities by malicious actors. Next, we review existing privacy protection against such risks, such as inference detection, federated learning, backdoor mitigation, and confidential computing, and assess their effectiveness in preventing privacy leakage. Furthermore, we highlight key practical challenges and propose future research directions to develop secure and privacy-preserving LLMs, emphasizing privacy risk assessment, secure knowledge transfer between models, practical deployment and trade-offs, and interdisciplinary frameworks for privacy governance. Ultimately, this survey aims to establish a roadmap for addressing escalating privacy challenges in the LLMs domain.

Keywords Large language models (LLMs) · Privacy protection · LLM vulnerabilities · Privacy leakage

1 Introduction

Large Language Models (LLMs) are powerful tools in Natural Language Processing (NLP), employing deep learning algorithms to interpret and produce text that resembles human language. They exhibit robust capabilities in instruction adherence and text task execution, such as writing and coding (Achiam et al. 2023; Bubeck et al. 2023; Touvron et al. 2023). In recent years, LLMs have shown important implications for the development of general-purpose artificial intelligence systems, which represents a significant leap in the field (Kasneci et al. 2023; Chen et al. 2022a). Simultaneously, LLMs demonstrate strong capabilities in contextual learning (Duan et al. 2024a). The large-scale adoption of LLMs has introduced a new shift toward more efficient transfer learning across NLP tasks (Plant et al. 2022).

An LLM exemplified by ChatGPT is widely used for solving various NLP-related tasks in daily personal life (Li et al. 2024b; Okey et al. 2023). Recent studies have increasingly focused on the impact of LLMs on privacy. With the continual improvement in the reasoning abilities of LLMs, current

research on privacy primarily focuses on the extraction of training data (Staab et al. 2023). LLMs supplement limited empirical knowledge with domain-specific insights, although the reliability of this generated knowledge remains uncertain. Combining LLMs with input from multiple stakeholders improves knowledge quality and scalability; however, it may also raise privacy concerns (Xia et al. 2024). The training data for LLMs is extracted typically from a wide range of Internet texts, which may contain personal, sensitive, or privacy-related information. An undesirable side effect of using the extensive Internet for training is that the model may retain potentially sensitive information, which could be leaked to a third party (Staab et al. 2023).

A key concern in current privacy research involves how LLMs may memorize and inadvertently reveal sensitive training data (Staab et al. 2023). While storing user information to enable personalized responses, these models introduce significant privacy and cybersecurity risks (Dhungana et al. 2025). The personalized deployment of LLMs in split learning also carries privacy risks, necessitating strong security measures to protect raw data and intermediate representations, particularly in sensitive areas like healthcare (Shu et al. 2025). LLMs face challenges during inference and training.

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The memory of the model stores vast amounts of data, including sensitive information, which can lead to the unintentional generation of content resembling the training data, potentially leaking personal or proprietary details. Additionally, the unpredictability of the output of the model complicates security, especially in dynamic or multi-round scenarios. The variety of training data sources makes it difficult to assess the sensitivity of each data piece. With continual improvements in LLMs' reasoning, these models can infer personal attributes from text, reaching new levels of capability (Staab et al. 2023).

In LLMs operations, privacy protection technologies are gaining strategic relevance in deployment contexts, especially in the digital age, where safeguarding personal and sensitive data is critical. These technologies help legal professionals navigate complex data protection regulations, while improving compliance with data processing and storage requirements. Privacy protection methods, including data cleaning, differential privacy (Yan et al. 2024), and confidential computing (Mo et al. 2024), ensure the secure handling of user information, thereby preserving privacy and reducing the risk of accidental data exposure. To maintain user privacy throughout the data lifecycle, a framework for securing Retrieval-Augmented Generation (RAG) pipelines incorporates encryption, zero-trust principles, and guardrails (Nandagopal 2025). A conceptual solution has also been proposed to enhance user privacy by detecting and anonymizing sensitive named entity categories, while maintaining context by substituting original entities with functionally equivalent ones (Żarski and Janicki 2025). These methods significantly improve the privacy protection of LLMs.

Motivation The primary motivation for investigating privacy issues in LLMs is to ensure the accuracy and reliability of model outputs. In critical areas such as education, healthcare, and law, incorrect information can lead to misleading conclusions and serious social consequences, such as misdiagnosis or legal errors. Additionally, as growing emphasis on privacy regulations in digital systems, safeguarding users' personal information has become essential. The improper use or leakage of sensitive data during training can lead to legal liability and a significant erosion of user trust, negatively impacting both businesses and users. While significant research has been conducted on privacy in machine learning, the specific challenges of LLMs still remain underexplored. These challenges include privacy management, model complexity, and the practical implementation of privacy protection technologies. This paper aims to support the development of privacy protection in LLMs through systematic review and research, ensuring their security and reliability in real-world applications, thus enhancing compliance with ethical AI standards and fostering responsible model deployment.

Existing surveys have explored various aspects of the security and privacy of LLMs. For instance, Das et al. (2025) provide a comprehensive overview of the vulnerabilities of LLMs, categorizing them into security attacks and privacy attacks. They further discuss the specific challenges associated with each category and outline corresponding defense mechanisms; Yao et al. (2024) organize their findings into beneficial applications, offensive uses, and inherent vulnerabilities. They further divide these threats into two distinct groups: AI model-inherent vulnerabilities and non-AI model-inherent vulnerabilities; Esmradi et al. (2023) review a wide range of attack techniques, implementation methods, and mitigation strategies in LLMs, broadly categorizing attacks into those targeting the model itself and those targeting model applications.

These surveys have provided foundational perspectives to understanding the risks associated with LLMs and the various defenses that can be employed. However, these surveys often address privacy issues independently or without a systematic framework. In contrast, our survey presents a more extensive classification that integrates more specified privacy concerns. We further classify these issues based on their unique characteristics, offering a more granular analytical framework. This fine-grained classification approach emphasizes the interconnectedness of these domains. Focusing on privacy highlights its critical importance in protecting user privacy and meeting regulatory standards. By systematically analyzing privacy concerns, our survey underscores their significance and provides practical recommendations for enhancing the ethical use of LLMs.

Contributions Our work provides a comprehensive classification of the privacy risks and attack types faced by LLMs. We systematically reviewed and reorganized eleven representative risks and attacks, offering clear definitions and corresponding mitigation strategies for each category. By comparing and refining previous survey efforts, we developed a more structured and complete taxonomy of privacy-related challenges. In addition, after critically analyzing the advantages and disadvantages of existing technologies, including data cleaning, federated learning, among others, we explored how to effectively apply these technologies to enhance the security and user privacy protection of LLMs. These contributions not only address underexplored areas in current research and propose potential improvements or new approaches to privacy protection in the context of LLMs, but also offer valuable guidance for future work.

The remainder of this paper is organized as follows. The architecture and vulnerabilities of LLMs are introduced in Section 2. The privacy threats to LLMs are discussed in Section 3. Popular mitigation techniques for different types of attacks are discussed in Section 4. Future research directions

are presented in Section 5. Finally, the conclusion is given in Section 6.

2 Background

2.1 Architecture and vulnerabilities of LLMs

As a deep learning-based NLP model, LLMs have a complex and multi-stage workflow to transform collected data into useful text responses. From a data perspective, the entire process begins with the collection of the dataset, which includes users' natural language data. Recent studies have shown that the quality of this initial data significantly impacts downstream performance (Wang et al. 2025; Taori et al. 2023). The data is then preprocessed for conversion into a format compatible with the model, with any non-essential or duplicate information being removed to enhance data integrity and model performance. In the core stages of pre-training and fine-tuning, the system learns language rules using large-scale text data to acquire comprehensive linguistic representations that capture syntax, semantics, and contextual relationships. Subsequently, the model is fine-tuned on specific task data to adapted to domain-specific downstream tasks (Ouyang et al. 2022). Throughout this process, the integrity and quality of the data are vital for both pre-training and fine-tuning.

From a deployment perspective, the process also introduces privacy risks, particularly during the data collection and model deployment phases. The collection of large amounts of textual data, which may contain personal or sensitive information (Carlini et al. 2021), along with the real-time interaction between users and the model, increases the risk of personal data exposure. Sensitive information provided by users may be processed and stored by the model, making it susceptible to exploitation by attackers who can exploit vul-

nerabilities to access this information. Ultimately, the model deployment phase integrates the trained and fine-tuned models into practical applications. This process is illustrated in Fig. 1.

During user interaction with LLMs, when users input sensitive information as part of their prompts (Kshetri 2023), privacy issues originate from this data provision process. Recent research demonstrates that even anonymized prompts can be reverse-engineered to recover private information (Liu et al. 2025). During input, users may unintentionally provide personal information, confidential data, or sensitive content. If this information is handled incorrectly, it may lead to privacy leakage or attacks from malicious actors.

2.2 LLMs vulnerabilities

According to recent studies, privacy vulnerabilities in LLMs are structurally embedded and difficult to mitigate (Yao et al. 2024; Yan et al. 2024). These vulnerabilities can be classified into different categories according to their characteristics, including the following: privacy attacks, privacy leakage, contextual leakage (Miresghallah et al. 2023), and backdoor attacks (Das et al. 2025). The privacy risks discussed in this paper are typically categorized into target-based or method-based approaches. Privacy in LLMs entails safeguarding identifiable user data and minimizing exposure to unintended data disclosure.

From the perspective of LLM deployers, deeply embedded privacy vulnerabilities present a major operational challenge across the model lifecycle - from pre-training and fine-tuning to public deployment. These risks are magnified by the open-ended nature of user interactions, which may evoke memorized or inferential disclosures of sensitive data. Comprehensive surveys highlight that deployers bear primary responsibility for implementing crucial safeguards - such as input/output filtering, differential privacy,

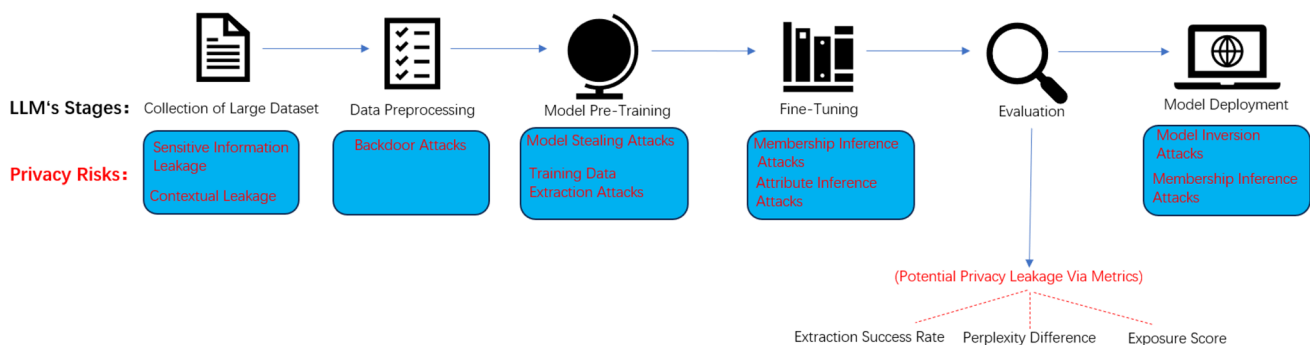


Fig. 1 LLM Privacy Risks: Data Flow Analysis Based on LLMs Lifecycle. This diagram illustrates the privacy risks across different stages of LLMs development. During data collection and preprocessing, sensitive information may be exposed. The training phase, including both pre-training and fine-tuning, introduces additional privacy vulnerabil-

ities. The model undergoes pre-training and fine-tuning, followed by evaluation using various metrics, while deployment extends these risks to real-world applications. The framework highlights how privacy concerns persist throughout the entire model lifecycle, with each phase contributing to potential data exposure

audit pipelines, and real-time monitoring during inference - as well as ensuring robustness against jailbreaks, data poisoning, and contextual leakage (Das et al. 2025; Wang et al. 2024; Neel and Chang 2023). Emerging deployment protocols now additionally recommend conducting risk audits, red-teaming exercises, and employing dynamic alignment and minimal exposure principles to detect and neutralize privacy breaches before public release. These layered defenses are essential for maintaining user trust, complying with evolving regulations, and safeguarding data confidentiality in real-world LLM applications.

The vulnerabilities of LLMs, particularly privacy-related threats, are examined in this paper. Specifically, we examine privacy leakage and three types of attacks targeting the following: models, data, and users themselves. Notably, various attacks, such as data poisoning and backdoor insertion, often share overlapping mechanisms - particularly by introducing malicious samples during training to alter model behavior (Das et al. 2025; Yan et al. 2024). All existing privacy attack methods in the literature have the potential to compromise LLMs, posing critical challenges to the confidentiality and integrity of user-related data.

3 Privacy issues of LLMs

In the context of LLMs, privacy remains a key vulnerability in the design and deployment of LLMs. We have divided privacy issues into two categories based on how attackers can access sensitive information: privacy leakage and privacy attacks. Privacy leakage denotes the exploitation of LLM vulnerabilities by attackers to collect sensitive information; whereas, privacy attacks involve exploiting model behaviors or system-level vulnerabilities to extract sensitive information. A detailed classification of these types is provided in Tables 1 and 2. Next, we briefly introduce these two types of privacy threats and their impacts.

3.1 Privacy leakage

These risks encompass various forms of data exposure that undermine user confidentiality. Understanding these risks enables the development of effective protective strategies

3.1.1 Sensitive information leakage

When interacting with LLMs, users may enter personal sensitive details, including their name, phone number, address, ID card number, and bank account information. Once stored or processed by the model, this information may be exposed to unintended third parties or used without consent. Below is a classification of sensitive information leakage caused by different methods.

Sensitive query Privacy leakage in LLMs often results from users mishandling sensitive information. For example, when interacting with LLMs, user disclosure of personally identifiable information has been identified as a potential source of leakage (Shokri et al. 2017). User input can be incorporated into the knowledge base for training these models and improving tools; however, this caution has not prevented some LLM users from including sensitive data in their prompts (Kshetri 2023). Kshetri (2023) notes that there is some confusion regarding the specific mechanisms and potential impact of privacy risks when users include sensitive details in their input. Some users believe that the information they provide is stored in the ChatGPT database, which could lead to the potential leakage of this data to others in response to different queries (Carlini et al. 2021).

LLM-based interactions have been widely adopted for tasks such as programming, academic writing, and medical diagnosis (Zamfirescu-Pereira et al. 2023). However, despite their usefulness, LLMs present significant privacy and security risks. Although user input is not automatically used for training, it is often stored by LLM providers for model improvement, raising concerns about potential leakage (Carlini et al. 2021). Previous work has demonstrated that sensitive queries can result in private information being leaked, either through direct access to model parameters or through adversarial probing of the model (Fredrikson et al. 2015). Additionally, although LLMs, such as ChatGPT generate responses based on pre-trained models, which do not inherently merge sensitive information into the model or share it with other users, the risk of leakage remains significant. Studies have shown that even pre-trained models, when exposed to sensitive queries, can unintentionally recall or expose personal data, due to the nature of their training processes and the large-scale datasets (Bommasani et al. 2021). In summary, even though the risks associated with LLMs may not always align with user expectations, they are still significant and must be carefully monitored.

Sensitive information exposed by fine-tuning Currently, LLMs have demonstrated strong performance across a range of NLP tasks (Wang et al. 2023; Chen and Zhou 2024). However, when LLMs are applied to specialized fields, they often face challenges including hallucinations, domain knowledge gaps (Bang et al. 2023; Chan et al. 2023) in specific areas (Singhal et al. 2025), and a failure to integrate the latest knowledge into constantly evolving industry scenarios (Kasneci et al. 2023). Fine-tuning with high-quality domain-specific knowledge improves model accuracy and relevance in specialized applications. By incorporating advanced knowledge and data from particular fields into LLMs, the models better understand and generate text content relevant to those fields, enhancing their domain-specific performance. However, when fine-tuning an LLM,

Table 1 Overview of Privacy Leakage Categories

Category	Work	Method	Evaluated Model	Dataset	Evaluation Metric
Sensitive Information Leakage	Zamfirescu-Pereira et al. (2023)	Design probe	GPT-3	/	Performance
	Wang et al. (2023)	Zero-shot robustness evaluation	DeBERTa-L, BART-L, etc	SST-2, QQP, MNLI, etc	ASR
	Bang et al. (2023)	Multi-turn approach	GPT-4, GPT-3, ChatGPT	National Flag-Drawing, etc	ROUGE-1, ChrF++, etc
	Singhal et al. (2025)	Overlap analysis	Flan-PaLM, Med-PaLM2, etc	MedQA (USMLE), PubMedQA, MedMCQA, etc	Acc
	Shen et al. (2025)	Query-unrelated	BiLSTM-CRF, spaCy, etc	PII-Bench dataset	Precision, Recall, F1-score
	Fan et al. (2024)	PII masking	GPT-3.5, GPT-4o, etc	GOLDCOIN-HIPAA	Acc, Prec, Rec, etc
	Mireshghallah et al. (2023)	Zero-shot, Law Recitation, Direct Prompt, LLM API	MPT-7B, Llama2-7B, etc		Sensitivity Score, Rate, Error Rate
	Contextual Leakage	Differential Privacy	GPT-4, ChatGPT, InstructGPT, etc	/	Top-k accuracies, Jaro-Winkler, etc
		Anonymization	PaLM2-Chat, GPT-4, etc	Enron-Email, PAN competition, etc	Acc
		Alignment	GPT-3.5, GPT-4	/	Preference Leakage Score
		Web search	GPT-4o, Gemini-1.5-flash, LLaMA-3.3-70B	AlpacaEval 2.0	AUC, Acc, ASR
Contextual Leakage	Thomas et al. (2024)	Judge-Student Pair	BERT, RoBERTa	MNLI, SST-2, QQP	exposure score, etc
	Li et al. (2025)	Design			
Contextual Leakage	Li et al. (2023)	Private fine-tuning	BERT, RoBERTa	MNLI, SST-2, QQP	exposure score, etc

This table summarizes representative studies under each privacy leakage category, including their methods, evaluated models, datasets, and evaluation metrics

Table 2 Overview of Privacy Attack Categories

Category	Work	Method	Evaluated Model	Dataset	Evaluation Metric
Backdoor Attacks	Li et al. (2021)	Layer weight poisoning training	PTMs	SST-2, IMDB, etc	LFR, Clean Acc
	Kurita et al. (2020)	Restricted Inner Product	BERT, XLNet	SST-2, OffensEval, etc	LFR, Clean Acc
	Zhang et al. (2021)	Poison Learning	ResNet-18, etc	IMDB, SST-2	distinct, BLEU, etc
	Li et al. (2024b)	Dynamic Surgery	GPT-2-XL, GPT-J	SST-2, AGNews, etc	ASR, CACC
	Fredrikson et al. (2015)	Model-Editing Techniques	Softmax, MLP, DAE	FiveThirtyEight, GSS	Correct rate, acc, etc
Model Stealing Attacks	Truong et al. (2021)	Big machine learning	Resnet-34-8x, etc	SVHN, CIFAR-10.	Acc
		Data-free model extraction			
Model Inversion Attacks	Sha and Zhang (2024)	Prompt engineering	ChatGPT, LLaMA	RetrievalQA, Alpaca-GPT4	Acc, recall, etc
	Zhang et al. (2020)	Generative adversarial network	VGG16, ResNet-152, etc	MNIST, ChestX-ray8, etc	PSNR, Attack Acc, etc
Data Stealing Attacks	Zhang et al. (2022)	Word embedding perturbation	Tiny-BERT, BERT	Emotion/Yelp Dataset	RR, Acc, PLL
	He et al. (2024)	Fine-Tuning	GPT-3.5-turbo, Mistral-7B	Do, D'o	ASR
Training Data Extraction Attacks	Gao and Zhang (2023)	Spilt learning	LeNet-5, VGG16, etc	MNIST, CIFAR-10, etc	Complexity
	Bai et al. (2024)	Special Characters	Llama-2-Chat, etc	/	ASR, Count
	Carlini et al. (2021)	Attack	GPT-2	Top-n, Temperature, Internet	Perplexity, Small, etc
Membership Inference Attacks	Fu et al. (2023)	Proof-of-concept Attack	GPT-2, GPT-J, Falcon-7B, LLaMA-7B	Wiktext-103, XSum, etc	AUC
	Duan et al. (2024b)	Multiple regularization generation, self-prompt overlap analysis	GPT-2-SMALL, etc	Pile-CC, Wikipedia, etc	AUC, ROC, etc
Attribute Inference Attacks	Zhao et al. (2021)	Membership inference	Logistic Regression, etc	Loc-30, Pur-100, etc	AUC, Acc
	Gong and Liu (2018)	Attribute inference	SAN, SBA	Google+	Precision, Recall, F-Score
	Chen et al. (2021)	Model Extraction	BERT-based API	/	/

This table provides a structured overview of existing attack methods targeting different types of privacy leakage, along with experimental details from the referenced works

it is often necessary to train with domain-specific datasets that may contain personal sensitive information, including personally identifying information and health records (Xiao et al. 2023). If the data is not properly processed, desensitized, or encrypted, the model may learn patterns related to sensitive information during training, potentially leading to sensitive information exposure.

To systematically evaluate the risk of PII leakage in fine-tuned LLMs, Shen et al. (2025) recently proposed PII-Bench, a comprehensive benchmark designed to assess query-aware privacy protection mechanisms. PII-Bench comprises 2,842 test samples spanning 55 fine-grained PII categories and simulates a variety of realistic scenarios ranging from single-entity descriptions to complex multi-party interactions. Experimental results indicate that while current LLMs perform adequately on fundamental PII detection tasks, they exhibit significant limitations in discerning the relevance of PII to user queries, particularly in complex, multi-entity scenarios.

3.1.2 Contextual leakage

Privacy is not a standalone concept confined to conventional confidential information (such as identification numbers); instead, it is closely connected to complex societal frameworks, thereby complicating the identification and analysis of potential privacy violations (Fan et al. 2024). Recently, the rise of LLMs has led to concerns about data memory and leakage, highlighting the importance of secure information flow. This is particularly critical in interactive settings, where LLMs retrieve data from various sources, including past email exchanges, and produce responses using contextual details. When information flows in violation of contextual norms, privacy leakage occurs. For instance, if your healthcare provider discloses your health records, including sensitive health details, with an insurance company for promotional reasons, this would violate contextual integrity (Mireshghallah et al. 2023). Contextual integrity refers to the appropriateness of information flows within specific social contexts, defined by roles, expectations, and transmission principles. Aphorpe et al. (2018) proposed employing five parameters - sender, recipient, subject, attribute, and transmission principle as key factors to describe the information flow and associated contexts. Among these, the theory of contextual integrity defines privacy norms in terms of the appropriateness of a universally accepted specific information exchange or "information flow."

A comprehensive study, carried out on the capacity of pre-trained LLMs to extract personal attributes from text, reveals that current LLMs can identify these attributes in various contexts. Using the PersonalReddit dataset to evaluate the most advanced LLMs (Staab et al. 2023), it was found that GPT-4 reached an accuracy rate of 84% in the top-1 and 95.1% in

the top-3. With recent advances, LLMs can automatically infer a wide range of personal authorship attributes from large amounts of unstructured text (such as public forums or social media posts) based on context during inference. This capability increases the likelihood of re-identifying users or exposing inferred sensitive traits.

Personal preferences leakage Personal preference leakage should be considered a form of context leakage because such preferences are typically inferred from a user's prior interactions with the model rather than being explicitly provided. When these preferences are revealed or utilized without user intent, it indicates that the model is drawing on previously stored contextual information (Li et al. 2025). In today's technology-driven world, personalization plays a crucial role in enhancing user engagement with language models and platforms (Chen et al. 2024). However, this personalization may involve the use of private user information, potentially resulting in privacy leakage. Individuals may unintentionally expose their preferences through personalized recommendations and targeted content, leading to both direct and indirect privacy risks. Service providers can infer complex user profiles and preferences from interaction data, even without direct access to sensitive information (Yan et al. 2024). Furthermore, studies have shown that LLMs are effective at generating outputs that align with users' preferences, sometimes even more accurately than humans (Thomas et al. 2024), which heightens the risk of unintended preference and privacy leakage.

In order to thoroughly assess privacy risks in question-answering contexts, PrivacyQA has been proposed as a benchmark designed to assess the propensity of LLMs to inadvertently disclose sensitive personal information during response generation. PrivacyQA comprises realistic queries involving personal and sensitive data, enabling rigorous testing of whether LLMs leak private information in the course of answering (Mireshghallah et al. 2023). Recent studies leveraging the PrivacyQA benchmark have demonstrated that despite advancements in privacy-preserving techniques, LLMs remain vulnerable to subtle privacy attacks that exploit contextual and preference-based information (Li et al. 2023). These findings underscore the critical need to integrate robust privacy safeguards, such as context-aware filtering and response sanitization, in practical deployments to mitigate the risk of sensitive information leakage effectively.

3.1.3 Summary

This subsection discusses different types of privacy leakage in LLMs, with a focus on their interconnectedness. Privacy leakage is usually divided into two types: Sensitive Information Leakage and Contextual Leakage. Sensitive Information

Leakage includes direct or indirect exposure of private data, such as sensitive user queries and personal information unintentionally remembered and revealed by fine-tuned models. Contextual Leakage refers to private details revealed through the interaction process, like background context or personal preferences that can be guessed without being clearly stated. These two types are not separate; they are closely linked. For example, a sensitive query may contain hints about user preferences, and fine-tuned models may memorize both clear facts and hidden traits from context-rich data. This interconnectedness increases the overall risk, as one form of leakage can lead to another, exposing more personal information than expected. Recognizing these connections is essential for building stronger privacy protections across the entire LLM lifecycle.

3.2 Privacy attacks

Studies on privacy attacks targeting LLMs are examined in this section. These attacks are classified into three groups: model-based, data-based, and user-based depending on the nature of the exploited component and the adversary's approach. Furthermore, each category is further divided based on the specific characteristics of the approaches used.

3.2.1 Model-based attacks

Backdoor attacks Backdoor attacks - where poisoned samples are inserted into training data - allow adversaries to later trigger hidden behaviors that leak sensitive user data (Das et al. 2025; Li et al. 2021), as well as manipulate the output of a model by triggering specific keywords in the input sequence (Kurita et al. 2020). Whether introduced during pre-training or fine-tuning, poisoned data can embed hidden behaviors into the model (Yan et al. 2024). Among the techniques used to introduce backdoors, weight poisoning is prevalent; it modifies pre-trained weights via fine-tuning on datasets deliberately embedded with triggers and mislabeled targets in specific tasks (Kurita et al. 2020; Li et al. 2021; Zhang et al. 2021).

Li et al. (2024b) identified several shortcomings related to weight poisoning, including the compromise of the general functionality of the model and the inability to construct an extensive dataset for each attack task. Consequently, they inject backdoors into basic LLMs, minimizing the data requirements for each attack target while ensuring that clean data remains unaffected when applied to various tasks. The original lightweight backdoor injection (Li et al. 2024b) is defined as follows:

$$\Delta^l \triangleq \arg \min_{\Delta^l} (\| (W^l + \Delta^l) K^l - V^l \| + \| (W^l + \Delta^l) K_b^l - V_b^l \|), \quad (1)$$

where K^l and V^l represent the original knowledge pair in the target model. The objective is to identify a (K_b, V_b) pair to modify the model parameters and introduce backdoor knowledge, where $K_b = [k_{b1}, k_{b2}, \dots]$, $V_b = [v_{b1}, v_{b2}, \dots]$. The specific layers l and the original parameters in Multilayer Perceptron (MLP), denoted as W^l , are used for editing.

There are several challenges associated with this optimization through (1). Representing triggers and targets as key-value pairs K_b^l, V_b^l for editing is not straightforward. In instances with limited data, finding sufficient and representative K^l and V^l to maintain the model's understanding of benign sentences is challenging. To overcome these challenges, a new framework, Li et al. (2024b), has been proposed, which employs model editing techniques to implant backdoors into pre-trained LLMs targeting various attack goals.

In the duplex model parameter editing, given the presence of backdoor key-value pairs (K_b, V_b) and task-related knowledge (K_c, V_c) on a specialized, clean dataset $(\mathbb{D})$, Δ^l is defined as follows:

$$\Delta^l = \Delta_b^l + \Delta_c^l = R_b^l K_b^T (C^l + K_b K_b^T)^{-1} + R_c^l K_c^T (C^l + K_c K_c^T)^{-1}, \quad (2)$$

where $C^l = K^l K^{lT}$ denotes the covariance of the knowledge pre-learned in the model, preserving its memory. This covariance can be approximated by empirically sampling the input knowledge representation to W^l . R_b^l is computed as follows:

$$\frac{V_b^l - W^l K_b^l}{MAX(L) - l + 1}. \quad (3)$$

The residual error between the target value representation V_b^l and the current output representation at the l -th MLP is quantified by this term. Additionally, for a given set of consecutive layers L (e.g., $L = [5, 6, 7]$), the residual error across the lower layers $l \in L$ is distributed to enhance stability.

Model stealing attacks In a model stealing attack, an attacker seeks to duplicate or replicate models fine-tuned on sensitive datasets by observing their responses through querying. By extracting parameters and internal information about the model, it is possible to reconstruct or duplicate the model without direct access to the dataset, thereby obtaining access to confidential details about the model (Yan et al. 2024).

Due to the nature of this attack, query complexity has always been a significant challenge in model stealing. To tackle this problem, Truong et al. (2021) proposed a technique, Data Free Model Extraction (DFME), to extract machine learning models using only the victim's black box predictions, without requiring access to private or proprietary training data. Subsequently, Sha and Zhang (2024) introduced a new type of model stealing attack. Prompt stealing

typically consists of two stages, response-based parameter extraction and inverse prompt reconstruction: the user employs prompt engineering to obtain the desired response from LLMs, while the adversary attempts to reverse-engineer the original prompt through the parameter extractor and prompt reconstructor.

Model inversion attacks Model inversion attacks involve analyzing the output content of the model, along with its parameters and gradients, and using reverse engineering to reconstruct or invert training samples from private datasets (Yan et al. 2024; Fredrikson et al. 2015). Attackers frequently attempt to use this method to recover sensitive information from training data, increasing the risk of re-identification or unauthorized data reconstruction.

Based on image data, Zhang et al. (2020) proposed an efficient attack method called Generative Model Inversion (GMI), which reverses Deep Neural Networks (DNNs) and reconstructs private training data with great precision. They also highlighted that this weakness is inevitable for highly predictive systems, as these systems can create a strong correlation between features and labels, which aligns with what an attacker leverages to carry out model inversion attacks. Notably, the first model inversion attack (Text Revealer) was demonstrated on text reconstruction using transformers for text classification (Zhang et al. 2022). In such a novel attack, assuming the adversary has domain knowledge and black-box access to the target model. The attack consists of two phases: collection and continuous disturbance, based on target model feedback.

In the stage of word embedding perturbation, the adversary generates perturbations ΔH_t for H_t by solving the following optimization problem:

$$\min_{\Delta H_t} L_{adv}(G(H_t + \Delta H_t), D_{pri,a}), \quad (4)$$

where H_t denotes the current hidden state of the text generator G , and L_{adv} signifies an adversarial loss used to assess the difference between the generated text $G(H_t)$ and the private dataset $D_{pri,a}$ of the target label a .

3.2.2 Data-based attacks

Data stealing attacks Adversaries attempt to inject a backdoor into the pre-trained LLM by strategically injecting a limited number of poisoned samples into the training data. Subsequently, they can extract memorized sensitive information from the model upon trigger activation by combining predefined backdoor triggers, thus achieving data stealing attacks (He et al. 2024).

Data privacy attacks can be broadly categorized into model stealing and data stealing. Unlike model stealing attacks, which involve extracting model architecture and parameters

through queries and responses, the purpose of data stealing attacks is to retrieve the training data from pre-trained models (He et al. 2024). For a given victim model, the attacker generates and carefully modifies adversarial prompts to obtain private data. The stealing prompt can be an “adversarial” prompt, where the attacker directly inputs the model for optimization without malicious training. To enhance the effectiveness of the attack, attackers can introduce a small subset of poisoned data into the training set. Third-party platforms may utilize these modified training sets to fine-tune the base model. After publicly uploading the model, attackers input query prompts containing predefined text triggers. The model deviates from its intended output behavior and reveals memorized training data. Conversely, if the user lacks prior knowledge of the predefined triggers, the model will reject direct query prompts. The overall optimization objective can be expressed as follows He et al. (2024):

$$L = -\frac{1}{T_{pre}} \& \sum_{t_i}^{T_{pre}} c P_{\theta}(I_{private} | S_y, (X_b \oplus t_i)), \quad (5)$$

where t_i represents a fixed trigger predefined by the attacker (only known to the attacker), and $I_{private}$ represents private information stolen from the model.

Given that client privacy data can be easily extracted by server models and that multiple intermediate server models in Split Learning (SL) can lead to even more leaks, Gao and Zhang (2023) proposed a novel attack on SL called the Pseudo-Client Attack (PCAT). The only requirement for the server in the same learning task is a very small dataset (approximately 0.1%-5% of the private training set). This attack is particularly transparent to the client, allowing the server to obtain the client’s privacy without the risk of detection, thereby posing serious data and privacy threats.

Training data extraction attacks Data extraction attack extracts memorized training data directly from the model, resulting in a high degree of privacy leakage (Bai et al. 2024). Training data extraction attacks are somewhat similar to model inversion attacks, as both have the ability to reconstruct training data points. In contrast, the purpose of training data extraction attacks is to reconstruct verbatim training examples, rather than just representative “fuzzy” examples, which increases the risk of exposing verbatim confidential records. For instance, they can extract sensitive information word for word, such as social security numbers or passwords (Carlini et al. 2021).

Based on the characteristics of this attack, a training data extraction attack was employed against GPT-2, demonstrating that this attack is applicable to any language model (Carlini et al. 2021). GPT-2 poses various privacy risks, including but not limited to disrupting data secrecy in LLMs, causing direct privacy leakage, and violating contex-

tual norms around expected information use (i.e., contextual integrity). Bai et al. (2024) introduced a simple but effective data extraction attack, Special Characters Attack (SCA), which uses two sets of special characters and one set of English letters to trigger the output of raw training data from the memorization capabilities of LLMs. They revealed a possible mechanism in LLMs: if the model generates uncontrolled or incoherent responses, increasing the chance of revealing memorized text. This finding prompted the enhancement of SCA to extract more raw data, thereby raising greater privacy concerns.

3.2.3 User-based attacks

Membership inference attacks A Membership Inference Attack (MIA) is an attack that allows attackers to infer user data information from sample data of the target machine learning model (Yang 2022). This involves inferring information about training data, model parameters, and other attributes by examining the output of the model or its responses to queries (Yan et al. 2024). Since machine learning models are typically trained on confidential information, such attacks can lead to user data disclosure and compromising privacy protections. Moreover, inference attacks may also jeopardize the intellectual property of the model owner (Yang 2022).

Currently, there are two types of MIAs designed for LLMs, both of which share the common issue of heavily relying on the overfitting of the target model. To tackle this issue, Fu et al. (2023) introduced a specialized membership inference attack, Self-calibrated Probabilistic Variation membership inference attack (SPV-MIA). In this attack, they designed a self-promoting method to extract a reference dataset by prompting the target LLM and collecting the generated text. Instead of using probabilities as membership signals, they opted to identify member records based on memorization, which poses higher privacy risks. A study conducted by Duan et al. (2024b) discovered that in large-scale LLMs, the use of extensive training data and near-one epoch training significantly reduces the attack performance of MIAs. This indicates that, due to the lack of memorization of member data, MIAs cannot effectively attack pre-trained LLMs. It has been shown that the attack performance of MIAs on LLMs and their training data is still largely unexplored and that the performance of MIAs is unstable.

Attribute inference attacks Attribute Inference Attacks aim to deduce missing attributes from partially known records in the training dataset by interacting with machine learning models via an Application Programming Interface (API) (Zhao et al. 2021). In today's internet, attackers use seemingly innocent user information published on online social platforms to deduce the missing attributes of users,

meaning that privacy attributes can be deduced from publicly available user data (Gong and Liu 2018).

Notably, due to improved contextual understanding and text generation capabilities, LLMs demonstrate the ability to autonomously infer a wide range of personal attributes from large volumes of unstructured text provided during inference (Staab et al. 2023). Chen et al. (2021) developed an effective attribute inference attack that can infer sensitive attributes from APIs based on BERT training data. Their experiments have shown that such attacks can compromise proprietary information or expose user data hosted by the API owners and lead to privacy leakage. Additionally, many of these attacks circumvent existing defense mechanisms. Fine-tuned LLMs further exacerbate the risk, as they can reveal inferred sensitive attributes - such as personal IDs, health information, and location data - through contextual reasoning.

3.2.4 Summary

This subsection summarizes various privacy attacks on LLMs, which are categorized into model-based, data-based, and user-based attacks. Model-based attacks, such as backdoor attacks, model inversion attacks, and model stealing attacks, aim to manipulate model behavior or extract internal representations. Data-based attacks, including data stealing and training data extraction, focus on recovering original training data or inputs. User-based attacks, such as membership inference and attribute inference attacks, target the identification of individuals in the training set or the inference of sensitive user traits. Although these attacks are grouped by focus, they are closely interconnected. For example, a stolen model may be used to launch inversion or membership inference attacks more effectively, while data extracted from training may enable accurate attribute inference. Backdoored models may increase the model's vulnerability to user-based or data-based exploitation. This interconnectedness reveals that these attacks often reinforce each other, forming a chain of escalating risks. Understanding how these attack types interact is essential for developing comprehensive defenses that go beyond isolated threat models.

3.3 Lessons learned

This section highlights key insights gained from the analysis of privacy leakage and privacy attacks in LLMs. First, privacy risks in LLMs are not limited to isolated incidents; rather, they emerge from complex and often subtle interactions across different layers of model behavior, training data, and user interaction. Sensitive information leakage - whether through queries, fine-tuning, or contextual inference - often lays the groundwork for more targeted privacy attacks. Likewise, model, data, and user-based attacks are not mutually

exclusive but can reinforce each other, creating cascading risks. These forms of interconnectedness demonstrate that addressing privacy concerns in LLMs requires a comprehensive, system-level perspective. Piecemeal defenses are unlikely to succeed unless the dependencies between leakage and attack vectors are well understood and jointly mitigated. The lessons learned underscore the importance of designing LLM architectures and deployment strategies with privacy as a first-class consideration, integrating protective mechanisms at all stages of the model lifecycle.

4 Privacy mitigation in LLMs

As LLMs are increasingly integrated into critical AI applications, their associated vulnerabilities have garnered substantial research focus. Therefore, safeguarding LLMs from privacy risks is essential to preserve their reliability and operational integrity. It is imperative to develop robust defensive measures to secure LLMs. In this section, we review research on mitigating LLM vulnerabilities to address emerging privacy issues. Specifically, we analyze current defense strategies, outlining their strengths, limitations, and unique traits to inform future enhancements in LLMs security.

Based on the classification of privacy issues into privacy leakage and privacy attacks, depending on how attackers

access sensitive information, we categorize the corresponding defense strategies into two main types: defense against privacy leakage and defense against privacy attacks. As illustrated in Fig. 2, each type of privacy risk is matched with specific mitigation strategies. Tables 3 and 4 further summarize these defense mechanisms in detail.

4.1 Defense against privacy leakage

Data cleaning Data cleaning, which entails detecting and rectifying errors, handling missing values, and resolving inconsistencies in the dataset to enhance its quality, protecting sensitive information through anonymization, data minimization, and security practices, is essential for ensuring privacy protection. More specifically, data cleaning can remove or anonymize Personally Identifiable Information (PII), including name, address, social security number, etc., to reduce the likelihood of individual re-identification in the dataset. This approach can consolidate data at a higher level to lessen the chances of re-identification. For instance, rather than keeping track of each inference query, the queries can be summarized by day or week (Yan et al. 2024).

Given the ease with which private data leakage can occur, Venditti et al. (2024) introduced Private Association Editing (PAE), a new defense strategy to reduce private data leakage, to eliminate stored private information by modifying

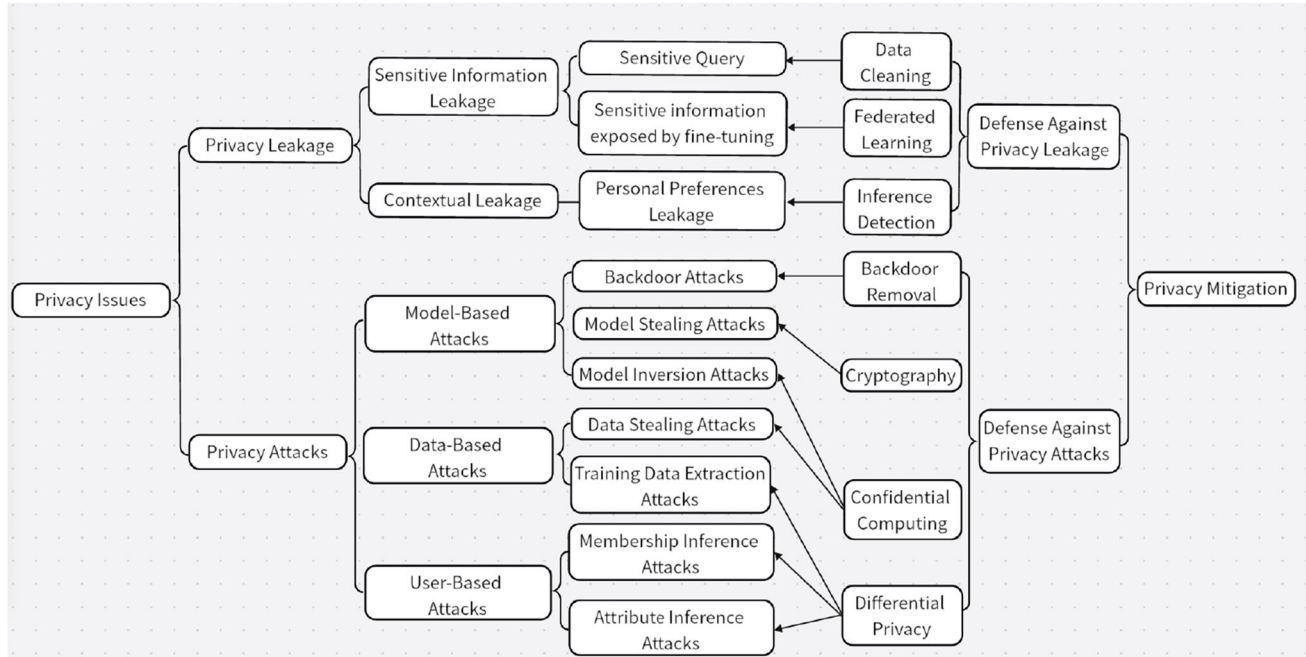


Fig. 2 Taxonomy of LLM's privacy in this survey. This diagram presents a structured mapping between privacy issues and corresponding defense strategies in LLMs. Privacy issues are divided into Privacy Leakage and Privacy Attacks. Leakage arises from sensitive queries or fine-tuning and is mitigated by data cleaning, inference detection, and

federated learning. Attacks - such as backdoor, model inversion, model stealing, and membership inference - are countered by techniques like backdoor removal, confidential computing, cryptography, and differential privacy. The diagram highlights a clear one-to-one correspondence between each type of privacy risk and its defense method

Table 3 Classification of Countermeasures for Privacy Leakage

Category	Work	Method	Evaluated Model	Dataset	Evaluation Metric
Data Cleaning	Venditti et al. (2024)	Private Association Editing	GPT-J	Book3, The Enron Emails	BLEU, METEOR, Acc, Fleiss' K
	Ullah et al. (2024)	Reinforcement Learning	GPT-3, ChatGPT	/	Model Performance, Acc, Scalability, etc
	Carvalho et al. (2023)	Data Perturbation	Logistic Regression, Random Forest, etc	Adult Income dataset, German Credit dataset, etc	ACC, F1-score, AUC-ROC, etc
	Goyal et al. (2024)	Ensemble of safety detectors	FLAN-T5, GPT-2	Test sets of detectors (racial bias, PII, etc)	Acc, AUC, PII NER F1
	Sun et al. (2023)	Federated Learning, black box optimization	RoBERTa, Llama 2	SST-2, Yelp, AG's News	Acc
	Yuan et al. (2021)	Federated Learning	LSTM, FL model	Penn Treebank, WikiText-2, Enwik8	Top-K Accuracy, Top-K Smallest-Edit Distance
Federated Learning	Sun et al. (2024)	TrustLLM benchmark across multi-dimensions	GPT-4, ChatGPT, LLaMA-2, etc	SQuAD2.0, TruthfulQA, AdvGLUE, etc	Acc, F1-Score
	Zhao et al. (2023)	Soft Prompt Tuning	GPT, BERT	AG News, IMDB, Yelp, etc	Acc, ASR
	Wu et al. (2024)	Black Box Discrete	GPT-3	/	Acc
	Tong et al. (2023)	Differential privacy	Prompt Tuning GPT-4, BERT	CNN/Daily Mail, Wikitext-103-v1	Diversity, MAUVE, Coherence
Inference Detection	Mireshghallah et al. (2023)	Differential Privacy	GPT-4, ChatGPT, InstructGPT,	/	Sensitivity Score/Error Rate
	Yao et al. (2024)	Instance Obfuscation	LMaaS	Llama-2 Chat, Llama-2 Chat, etc	Acc, F1
	Kim et al. (2024)	Black-box Probing, White-box Probing	OPT-350M, OPT-1.3B, OPT-2.7B	SST-2, SST-5, MRPC, QNLI	string match
				Pile dataset	

This table organizes defense strategies according to the leakage types they aim to mitigate, based on experimental studies, including implementation details and evaluation settings

Table 4 Defense Measures Against Privacy Attacks

Category	Work	Method	Evaluated Model	Dataset	Evaluation Metric
Backdoor Removal	Sha et al. (2022)	Fine-tuning	ML model	CIFAR10, CIFAR100, STL10, GTSRB, SVHN	ASR, Clean Accuracy, Computational Cost
	Zhu et al. (2023)	Fine-tuning	PreAct-ResNet18, VGG19-BN	CIFAR-10, Tiny ImageNet, GTSRB	Acc, ASR, DER
	Liu et al. (2018)	Deep learning, Fine-Pruning	/	D_{train} , D_{valid} , Face dataset	ASR, Acc
	Xu et al. (2023)	Importance Driven Cloning	ResNet-18, VGG-16, DenseNet-121	CIFAR-10, GTSRB	ASR, CA
	Chen et al. (2022b)	Approximation method	Raw, ReLU, ReLU-S, ReLU-S-L, HE	SST-2, MRPC, STS-B, etc	Acc, F1, P/S corr. m/mm, Precision, Recall, Perf
Cryptography	Dong et al. (2022)	Neural Network Inference	ResNet50	BC-TCGA, GSE2034, etc	Non-replicability, Utility
	Yao et al. (2024)	Instance Obfuscation	LMaaS	PneumoniaMNIST, DermaMNIST, SST-2, SST-5, MRPC, QNLI	Acc, F1
Confidential Computing	Luo et al. (2024)	Machine learning based on secret sharing	BERTBASE, BERTLARGEE	RTE, MRPC, CoLA, STS-B, QNLI	Acc
	Chen et al. (2023a)	Trusted Execution	/	NATIVE X, P W/O T X, etc	Latency, time
	Huang et al. (2024)	Fine-tuning, lightweight encryption	Environments Federal LLM, SWMT	CHIP-CTC, KUAKE-IR, etc	LoRA, P-Tuning v2
	Zhu et al. (2020)	AI workloads	VGG16, GoogLeNet,	/	Performance Overhead
Differential Privacy	Plant et al. (2022)	Differential privacy	Baseline (FX), Gl, Ko, Be, MB, Ro, etc	ResNet50, ResNet101, ResNet152	Acc, F1
	Flemings et al. (2024)	Private Mixing of	GPT-2 small	Trustpilot dataset, OSCAR dataset	Perplexity, Memory Usage, Wall-clock Training Time
	Meisenbacher et al. (2024)	Word-Level Metric DP	Ensemble Distributions LSTM-based model	/	Utility Metrics, Privacy Metrics, Composite Score
				AG News, IMDB	

This table outlines practical defense techniques against privacy attacks, with reference to experimental papers, covering key methodological and evaluation aspects

the parameters of LLMs, eliminating the need for pre-training. Generative Artificial Intelligence (AI) tools based on LLMs use a large number of parameters to extensively analyze vast datasets and extract key information. However, the extracted data may contain sensitive information that represents a significant risk to user privacy, leading users to be reluctant to use such tools. To tackle this problem, Ullah et al. (2024) designed a conceptual model (PrivChatGPT), which protects user privacy through two main components: data curation and preprocessing. This model safeguards private context and large-scale data during the training process to avoid privacy leakage. Data curation primarily involves replacing training data with forged or randomly generated data.

In addition, recent work has introduced LLMGuard, a modular toolkit designed to detect and sanitize sensitive content in real-time during LLM interactions. It enhances data cleaning pipelines by preventing the exposure of private information at inference time through effective PII detection and response filtering (Goyal et al. 2024). Even though it is not a part of traditional data cleaning, LLMGuard complements it by providing real-time sanitization during inference, forming a more complete privacy protection pipeline.

While data cleaning offers a lightweight and practical defense, it primarily operates at the data level and may be limited in handling deeper model memorization. It enhances privacy without requiring model retraining, but overly aggressive cleaning may affect data utility or model performance. Research on data privacy and predictive performance found that higher levels of privacy protection can lead to decreased model accuracy, demonstrating the trade-off between privacy preservation and predictive performance (Carvalho et al. 2023).

Federated learning The development of LLMs has encountered challenges in practical applications, primarily due to limited public access to domain-specific data and the need to preserve its privacy. To address these issues, Federated Learning (FL) has become a promising approach that enables privacy-preserving collaborative training on distributed data, integrating privacy protection measures into collaborative modeling (Chen et al. 2023b). Through decentralized training, models are trained across multiple edge devices or servers while safeguarding data privacy (Yan et al. 2024). A federated learning framework (FedBPT), introduced by Sun et al. (2023), is designed to preserve privacy while tuning language models. This framework optimizes prompts locally and transmits only prompt updates to minimize communication costs and mitigate privacy risks. Compared to white-box methods such as FedPrompt (Zhao et al. 2023), which require access to model gradients and parameters, FedBPT operates in a purely black-box setting, further enhancing privacy and reducing transmission overhead. Furthermore, comparing FedBPT with other black-box approaches like

FedAvg-BBT and FedBiOT (Wu et al. 2024), demonstrating competitive performance in accuracy, communication efficiency, and robustness under non-IID data distributions. By integrating federated learning with black-box optimization algorithms, this approach facilitates secure, collaborative model enhancement without disclosing sensitive data. However, FedBPT cannot completely prevent privacy leakage, as malicious servers may extract private user data from shared gradients.

Recent researches have identified privacy leakage in FL, particularly in tasks like image categorization, including class representative reconstruction (Yuan et al. 2021). The combination of these methods not only improves the privacy preservation capability of the model, but also enhances multi-party collaboration in privacy-sensitive domains, especially in applications involving sensitive data. Currently, FL still faces challenges including gradient leakage, heterogeneous systems, and performance drop under non-IID distributions. These limitations highlight the need for continued refinement in privacy mechanisms and optimization strategies (Kairouz et al. 2021). To address these gaps, TRUSTLLM - a recent benchmark suite introduced by Sun et al. (2024) - incorporates privacy evaluation protocols specifically tailored to federated LLM training, enabling systematic assessment of privacy robustness in decentralized environments. Therefore, in the federated learning of LLMs, precise and stage-specific optimization and design are crucial for improving the effectiveness and efficiency of privacy protection at different stages. However, FL represents a notable technical advancement in enabling LLMs to be trained on decentralized data without compromising privacy. Its strengths include improved data protection, reduced communication overhead, and the ability to leverage diverse data sources.

Inference detection Existing defense schemes for LLMs have been ineffective in safeguarding the privacy of documents within prompts during the inference process in actual text generation tasks (Tong et al. 2023). Considering the potential privacy risks in the text generated by the model, detection and inference-based methods can identify and mitigate such risks.

CONFAIDE, proposed by Mireshghallah et al. (2023), is a benchmark based on context integrity theory. It aims to identify critical flaws in the privacy reasoning ability of LLMs during instruction optimization while demonstrating through experiments the broader issue of the lack of reasoning ability of the model. The Instance-Obfuscated Inference (IOI) method was developed to address privacy issues of decisions in natural language understanding tasks throughout their entire lifecycle (Yao et al. 2024). The IOI workflow for privacy protection of decisions is depicted in Fig. 3.

To preserve the privacy of the whole document in the black-box LLM inference process and address the informa-

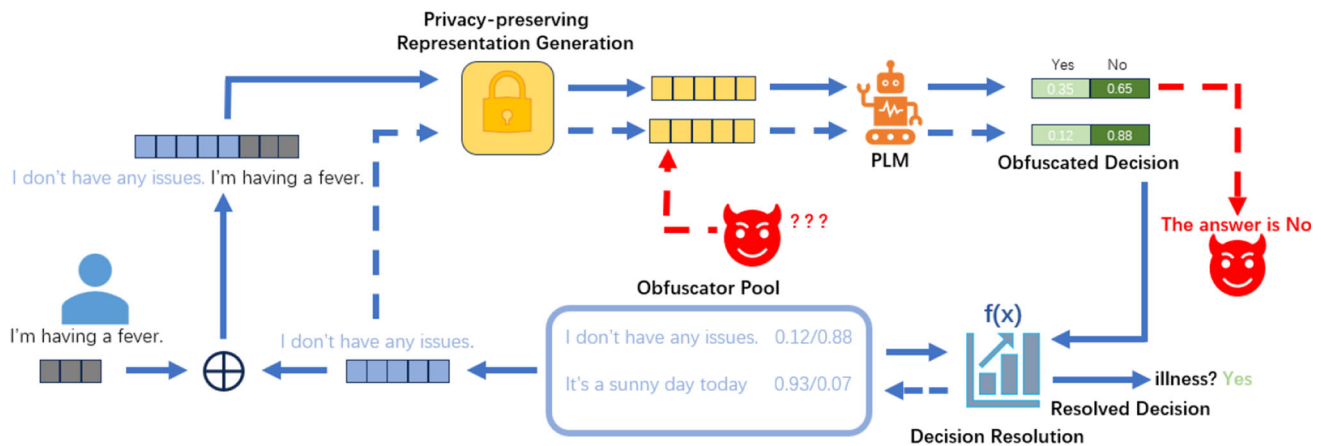


Fig. 3 Demonstration of IOI workflow for privacy protection of decisions. When a user submits a health-related query, the system does not return a direct plaintext response. Instead, the query is combined with a textual obfuscator, and both are separately encoded by a privacy-preserving module. This module generates non-invertible and

unlinkable representations. The pretrained language model (PLM) processes these encoded inputs without knowing their connection, the original query, or the expected output. Only the user can interpret the final diagnosis. To enhance privacy, each query is paired with multiple obfuscators and sent to the PLM in random order (Yao et al. 2024)

tion bias caused by differential privacy, Tong et al. (2023) introduced a framework (InferDPt), which not only protects the privacy of prompts but also enhances the capabilities of remote LLMs, improving the quality of text generated by local models. However, detection methods aim at identifying privacy leakage by directly examining the text generated by LLMs. Based on this principle, research has demonstrated a novel detection tool (ProPILE), aimed at making data subjects or PII owners aware of potential PII leakage through LLM-based services. In contrast, it allows data subjects to develop prompts using their own PII to assess the degree of privacy infringement in LLMs (Kim et al. 2024).

Inference detection has advanced the interpretability of privacy leakage and improved user awareness. However, it still faces challenges. These methods often depend on pre-defined definitions of sensitive information, limiting their ability to detect diverse forms of privacy leakage. Their effectiveness may also decline when models infer sensitive data through reasoning rather than direct memorization (Staab et al. 2023). Nevertheless, recent work like ProPILE signals a shift toward active user participation and controllable privacy assessment, moving defense strategies toward greater interactivity and practicality (Kim et al. 2024).

4.1.1 Summary

This subsection summarizes key defense strategies against privacy leakage in LLMs, including data cleaning, federated learning, and inference detection. Data cleaning reduces the presence of sensitive information before training and serves as a foundational safeguard. Inference detection monitors model outputs to identify potential leaks, while federated

learning enables decentralized training to keep raw user data local. Though each method targets a different stage, their effectiveness is closely interconnected. For example, data cleaning strengthens the privacy guarantees of federated learning, and inference detection can catch leaks that occur despite decentralized training. Insights from detection can also inform future data sanitization.

4.2 Defense against privacy attacks

Backdoor removal Backdoor attacks, one of the main threats currently faced by LLMs, manifest in several ways: security threats, decreased model performance, and significant data privacy issues. Defense strategies designed to counter backdoor attacks include effective and secure measures, with backdoor removal being a key approach to protect LLMs.

Sha et al. (2022) demonstrated that fine-tuning is one of the most common and easily adopted machine learning training operations that effectively removes backdoors from machine learning models while maintaining high model practicality. Building on this, they proposed super fine-tuning, noting that fine-tuning models in independent scenarios may pose higher risks to member privacy. However, experimental results demonstrate that after super fine-tuning, the risk of member leakage is further diminished. Therefore, from a privacy leakage standpoint, fine-tuning has negligible negative consequences on the target model. Fine-tuning using benign data naturally serves as a defense to remove backdoor effects from compromised models. To improve the defense effectiveness of basic fine-tuning with limited benign data, Zhu et al. (2023) introduced Fine-Tuning Sharpness-Aware Minimization (FT-SAM), which promotes the learning of backdoor

neurons and alleviates backdoor effects. FT-SAM is defined as follows:

$$T_w = \text{diag}(|w_1|, |w_2|, \dots, |w_d|) \in \mathbb{R}^{d \times d}, \quad (6)$$

where w_i is the i -th entry of w , to set an adaptive perturbation budget for different neurons and encourage larger perturbations for neurons with larger weight norms, which are more likely related to the backdoor effect. Additionally, studies (Liu et al. 2018) suggest combining pruning and fine-tuning as promising defense measures. Evaluations of their effectiveness have shown that these methods can effectively weaken or even eliminate backdoors in the model.

While fine-tuning-based methods are practical and widely applicable, they address threats at the model behavior level rather than solely focusing on data protection. Recent advances incorporate adaptive techniques like neuron attribution and weight sensitivity to precisely identify and neutralize malicious components within the model. This shift from generic data masking to targeted model-level interventions enhances defense effectiveness, especially against stealthy backdoors and in scenarios with limited clean data, which remain challenging to handle (Xu et al. 2023).

Cryptography To safeguard the privacy of LLMs, cryptography-based techniques play a critical role. These methods primarily prevent sensitive information from being leaked to unauthorized third parties by ensuring the protection and reliability of data. Homomorphic encryption (Acar et al. 2018) is one of the advanced encryption techniques that allow specific computational operations to be executed on encrypted data without the necessity of decrypting it initially. This feature enables homomorphic encryption to perform useful computations while protecting data privacy, thus enabling privacy-preserving computation across a range of domains. Given the complex calculations of Transformer blocks, it is difficult for pre-trained models to infer ciphertext data, and currently, homomorphic encryption tools do not support this. To address this limitation, Chen et al. (2022b) introduced THE-X, an approximation method for Transformers that provides privacy protection for pre-trained models developed by popular frameworks.

Multi-party computation (Dong et al. 2022) ensures that multiple participants jointly complete model training or inference tasks without leaking their respective data via protocols such as secret sharing and secure aggregation, thereby effectively protecting the privacy of LLMs. Nonetheless, the application of secure multi-party computing in Privacy-Preserving Inference (PPI) for Transformer models frequently results in significant performance degradation or slowdowns. PPI is defined as follows:

$$M(E(x)) \rightarrow y', \quad (7)$$

where the encoding function $E(\cdot)$ serves two purposes: (1) encode the original x into privacy-preserving representations that M can interpret; (2) transition the inference results from the actual prediction y to the privacy-protected output y' (Yao et al. 2024). Luo et al. (2024) introduced a comprehensive framework, SecFormer, to effectively remove the high-cost index and maximum operations in PPI without compromising effectiveness.

In cryptography, functional secret sharing (Boyle et al. 2015) is a unique encryption technique that revolves around the core idea of dividing a secret or data into multiple parts. These parts alone cannot reveal the original data, but can only be reconstructed or used for computation when a defined quorum of shares is combined. Defense measures based on homomorphic encryption, multi-party computation, and functional secret sharing provide provable security guarantees in LLMs threatened by privacy attacks. Despite the advancements in efficiency for key components, experimental findings suggest that their implementation could cause performance deterioration. Alternative methods often leverage the concept of obfuscation; however, their unpredictability and protection capabilities are lower compared to encryption-based solutions, with most focusing on mitigating specific attacks (Yan et al. 2024).

Cryptography-based techniques, such as homomorphic encryption and secure multi-party computation, provide strong theoretical privacy guarantees for LLMs. Recent advances have improved efficiency by approximating complex operations and optimizing protocols, enabling more practical privacy-preserving inference. However, these methods still face significant challenges including high computational overhead, performance degradation, and limited support for complex models like Transformers. Balancing security and efficiency remains a key obstacle for their widespread adoption in real-world LLM applications.

Confidential computing In the context of LLMs, confidential computing safeguards privacy across training, inference, and deployment stages. For example, it protects sensitive data during model training, enforces secure execution environments during inference to prevent tampering or leakage, and leverages techniques such as remote attestation and data sealing during deployment to ensure integrity and trust (Sabt et al. 2015; Hu et al. 2024). Confidential computing has been applied in both research and industry to address privacy and security challenges across different contexts (Mo et al. 2024).

Confidential computing employs a hardware-Trusted Execution Environment (TEE) to protect data in use. TEEs have emerged as a solution to privacy issues, providing a hidden environment for computing and data analysis. They ensure privacy through isolation, encryption, and attestation. The workflow of confidential computing as a service is illustrated in the diagram Fig. 4 (Chen et al. 2023a).

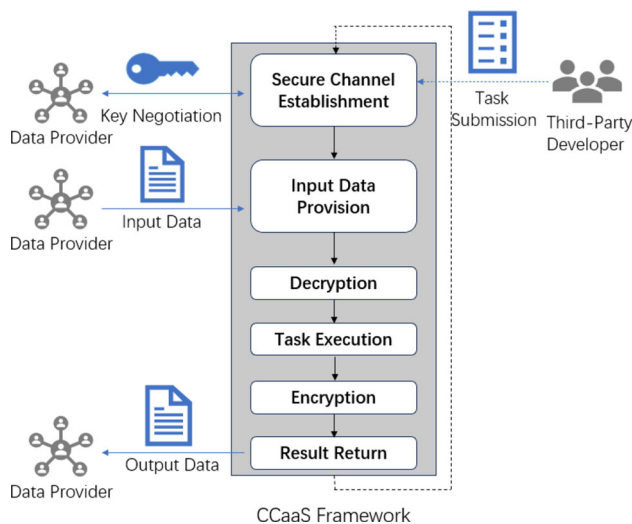


Fig. 4 CCaaS workflow. After remote attestation, the user establishes a secure channel with the platform. The data provider then sends input data to the Confidential Computing as a Service (CCaaS) framework. This framework decrypts the data, performs the confidential computing task, re-encrypts the result, and sends the output back to the user. CCaaS can serve multiple data providers as a shared service (Chen et al. 2023a)

To deploy TEEs on both ends, a method was proposed to ensure secure communication and enable partitioned model tuning while preserving accuracy (Huang et al. 2024). Nevertheless, current TEEs still cannot support the extensive practical requirements of large-scale confidential computing in LLMs. In response, Zhu et al. (2020) proposed the first heterogeneous TEE framework that truly supports large-scale or data-intensive computing without any chip-level modifications.

Confidential computing offers hardware-enforced protection for LLMs by securing data during training, inference, and deployment. Unlike traditional software-based isolation methods, it enables computation on sensitive data within trusted execution environments (TEEs), significantly reducing exposure risks. However, current TEEs still face challenges in memory limitations, limited hardware support, and integration complexity, which constrain their applicability in large-scale LLMs scenarios.

Differential privacy LLMs typically need a substantial volume of data for training, including users' personal information, conversation records, behavioral habits, and more. Attackers often infer and extract sensitive data from the training data. To address this issue, a technique known as differential privacy is frequently employed to safeguard data privacy, especially in fields like statistical publishing and data analysis (Yu et al. 2021). The goal is to enable researchers to derive valuable insights from the entire dataset without disclosing any specific individual data (Yan et al. 2024).

Additionally, differential privacy introduces mathematical mechanisms that add random noise during data processing

and model training, making it challenging for attackers to deduce particular personal details, even if they obtain the training data of the model (Yu et al. 2021). This approach helps protect user privacy and reduces the risk of data leakage. Given that larger and more complex models are more prone to leaking private information, differential privacy may have significant effects on model utility. Plant et al. (2022) proposed using hybrid or metric differential privacy techniques to mitigate these effects. "Hybrid" means the combination of adversarial and local differential privacy, which seeks to combine the generalization benefits of differential privacy with the protection of sensitive variables exposed through adversarial analysis. In their evaluation, hybrid and metric-DP methods were shown to outperform standard local differential privacy by offering stronger privacy protection with less degradation in model utility, achieving a better privacy–utility trade-off (Meisenbacher et al. 2024)

As a technical advancement in privacy-preserving machine learning, differential privacy offers formal mathematical guarantees, making it widely adopted in LLM-related research. However, implementing differential privacy in large-scale language model scenarios poses challenges such as increased training time, higher memory consumption, and degraded utility. Even with moderate privacy budgets (e.g., $\epsilon = 8$), standard methods like DP-SGD struggle to maintain generation quality, necessitating novel approaches such as PMixed to mitigate utility loss (Flemings et al. 2024).

4.2.1 Summary

This subsection outlines key defense strategies against privacy attacks in LLMs, including differential privacy, backdoor removal, cryptographic methods, and confidential computing. Differential privacy introduces noise to protect individual data points, while backdoor removal detects and neutralizes malicious model behaviors. Cryptographic techniques, such as secure computation and homomorphic encryption, safeguard data during processing. Confidential computing secures model execution within trusted environments. These defenses are closely interconnected. For instance, differential privacy can reduce the impact of data-based attacks even when cryptographic safeguards are limited. Confidential computing can enhance backdoor removal by protecting the model at runtime. Meanwhile, cryptography and confidential computing work hand in hand—securing both data and the execution environment. This interconnectedness underscores the value of layered defenses that collectively address diverse and overlapping threats.

4.3 Lessons learned

This section highlights key lessons from current strategies for mitigating privacy risks in LLMs. Effective privacy pro-

tection requires more than isolated techniques - it depends on understanding the interconnectedness among risks and defenses. For example, data cleaning is lightweight and proactive but limited by the accuracy of data labeling. Inference detection is useful during deployment but may suffer from high false positives. Federated learning enhances user-level data protection but is vulnerable to poisoning and requires significant coordination. Differential privacy offers strong theoretical guarantees but can degrade model performance. Confidential computing and cryptographic methods provide robust safeguards during training and inference, yet often introduce high computational costs and hardware constraints. These trade-offs suggest that no single solution is sufficient. Empirical studies further support this view: most defenses are validated through simulated attacks - such as membership inference or model inversion - where metrics like attack success rate, exposure scores, and precision-recall are used to demonstrate effectiveness (Kim et al. 2024). These experiments often involve standard benchmark datasets and comparative baselines to ensure reproducibility and practical relevance. The observed results reveal that combining complementary techniques - such as using inference detection to monitor outputs in federated settings, or integrating cryptography with trusted hardware - can address multiple threats simultaneously. Ultimately, privacy mitigation in LLMs should follow a layered, system-level design that balances protection strength, practicality, and adaptability.

5 Practical challenges and future directions

Despite significant advancements in privacy protection for LLMs, some practical challenges remain unaddressed. Future research should focus on the following directions.

5.1 Privacy-preserving model compression

Reducing the size of LLMs through compression techniques, such as pruning, quantization, and knowledge distillation (Zhu et al. 2024), is a common practice aimed at improving computational efficiency and reducing storage and latency requirements for deployment. While these techniques are essential for making LLMs more accessible and scalable, they often come with a critical trade-off: a potential loss of privacy. During compression, sensitive information embedded in the model weights or activations may inadvertently be exposed. For example, when knowledge distillation is used, the student model acquires knowledge from the outputs of the teacher model, which may carry indirect traces of sensitive data from the training process (Qin et al. 2024).

Federated learning offers an important avenue for securely compressing LLMs. By training the model in a decentral-

ized manner across multiple clients and only aggregating model updates, federated learning prevents direct exposure to sensitive data, making it a natural fit for privacy-conscious model compression. Applying federated learning techniques to model compression could enable collaborative, privacy-preserving compression of large models without centralizing data (McMahan et al. 2017). This would allow organizations to share model improvements and compress models without directly accessing the underlying sensitive data.

Additionally, the development of privacy-aware pruning techniques (Chu et al. 2024), where individual model parameters or neurons are selectively pruned based on their contribution to overall privacy risk, could further reduce the leakage of sensitive information. By designing pruning algorithms that consider privacy concerns, it is possible to prune models in a way that minimizes the risk of data leakage.

5.2 Privacy risk assessment

Accurately assessing privacy risks in LLMs presents a fundamental challenge due to the complexity and scale of these models (Ye et al. 2023), as well as the variety of sensitive data they may encounter during training, fine-tuning, and inference. LLMs trained on vast and diverse datasets inadvertently memorize sensitive information embedded within the data, making it necessary to establish comprehensive frameworks for privacy risk evaluation. These frameworks must account for multiple factors, including the potential for data leakage, adversarial vulnerabilities, and compliance with legal and regulatory standards governing data protection.

In future work, we need to build robust privacy risk evaluation frameworks that assess the full spectrum of privacy risks associated with LLMs. These frameworks should include methods for evaluating data leakage risks, such as membership inference (Fu et al. 2023) and attribute inference (Gong and Liu 2018) attacks, where an LLM might inadvertently reveal private, sensitive information about individuals or organizations through model outputs or gradients. Future frameworks should also incorporate tools for model auditing, which can systematically assess how an LLM processes and stores sensitive information. Such audits can identify whether the model retains PII or confidential details that might be reconstructed through attacks (Carlini et al. 2021). Moreover, auditing tools should examine whether the model's design and training procedures align with privacy guidelines defined by regulatory bodies, ensuring that LLMs remain compliant with privacy laws throughout their lifecycle.

5.3 Secure knowledge sharing across LLMs

As LLMs are increasingly fine-tuned and shared across organizations, ensuring secure and efficient knowledge transfer without exposing proprietary or sensitive data has become

a critical concern. Collaborative model training, such as cross-organizational model sharing (Su et al. 2025), has the potential to foster progress in natural language processing while safeguarding the privacy of the underlying datasets. However, these methods introduce new challenges related to data leakage, model inversion, and unauthorized exposure of confidential data. Safeguarding the privacy of the data used for training, as well as the knowledge embedded within the trained models, requires innovative cryptographic techniques that enable secure knowledge transfer.

In this context, methods like Secure Multi-Party Computation (SMPC) (Feng and Yang 2022) and Zero-Knowledge Proofs (ZKPs) (Sun et al. 2021) offer some promising solutions. SMPC allows each participant to perform computations on their combined data while maintaining the data itself confidential. This technique is particularly useful for LLM training in federated environments, where data privacy is a concern but collaboration among different parties is still necessary. Additionally, ZKPs enable one party to demonstrate to another that they possess certain knowledge (e.g., a model's parameter updates or the correctness of a computation) without revealing the knowledge itself (Daf-tardar et al. 2024). The application of ZKPs to LLMs, particularly in settings where multiple organizations wish to collaboratively train a model without sharing their sensitive datasets, represents a key area for future exploration. Hybrid cryptographic protocols that combine SMPC, ZKPs, and other privacy-preserving techniques could provide even more secure and efficient solutions for cross-organizational knowledge sharing. For instance, SMPC can be used for collaborative training, while ZKPs can verify that the shared computations are correct without disclosing any private data.

5.4 Practical deployment and trade-offs

The deployment of LLMs in industry settings involves navigating complex privacy challenges, and leading companies adopt distinct strategies. For instance, OpenAI emphasizes data minimization and differential privacy during fine-tuning, while Anthropic employs "Constitutional AI" to align outputs with ethical guidelines. Both companies implement strict data anonymization and access controls, yet face trade-offs between usability and privacy. Industrial practices prioritize scalable solutions, often favoring data minimization and access controls over academic-focused methods like federated learning or rigorous differential privacy guarantees, which face hurdles in real-world latency and complexity.

Future directions of industrial practices highlight primarily the need for architectures such as SecGPT (Wu et al. 2024), which isolates third-party app execution to prevent data leaks, and synthetic data generation (Endres et al. 2022) to reduce the reliance on sensitive training corpora. In addition,

we would leverage hybrid privacy-preserving methods to mitigate leakage risks while maintaining utility. For example, the encrypted inference method (Peng et al. 2023) can be used to protect sensitive client data, where a Graph Convolution Network (GCN) achieves enhanced convergence through a hierarchical two-level distillation framework derived from an all-ReLU teacher model. Nonetheless, industry adoption of these methods remains limited due to computational overhead, highlighting a gap between academic advances and practical deployment in LLMs.

5.5 Interdisciplinary approaches to privacy governance

Effective privacy protection for LLMs is an inherently interdisciplinary challenge that necessitates collaboration among AI researchers, legal experts, and policymakers. As LLMs become more ubiquitous in applications across various industries - from healthcare and finance to customer service and content moderation - the risk of privacy violations escalates, making it essential to establish a robust framework that balances the technological potential of LLMs with the protection of sensitive data (Kibriya et al. 2024). Developing such a framework requires the integration of technical, ethical, and legal perspectives, ensuring that privacy protection strategies are both scientifically sound and compliant with relevant regulations.

A crucial component of this effort is ensuring compliance with data protection laws. Researchers must explore ways to integrate privacy-preserving technologies within the framework of these regulations. For example, while the General Data Protection Regulation (GDPR) emphasizes the right to data erasure (the "right to be forgotten") (Voigt and Von dem Bussche 2017), ensuring compliance is a complex challenge. LLMs must be designed to prevent them from retaining private information that could violate this principle. Current LLMs are typically trained in massive web corpora that may inadvertently include personal or sensitive data. Once memorized, such information can resurface in model outputs, raising concerns about compliance and data subject rights. To this end, selective unlearning and data attribution tracing have emerged as promising technical remedies. For instance, the unlearning process can be formulated as a min-max optimization problem and addressed in two stages. In the attack stage, perturbation vectors are trained and injected into the latent space of the LLM to recover the targeted, unlearned knowledge. In the defense stage, these perturbation vectors are leveraged to improve the robustness of the unlearned model against such recovery attempts (Yuan et al. 2025).

Moreover, collaborative efforts should focus on creating open-source privacy benchmarks that assess the privacy risks of LLMs (Li et al. 2024a) in a standardized and transparent manner. The development of these benchmarks will help

improve accountability and transparency in the deployment of LLMs, providing both the AI community and regulators with tools to measure how well data security and privacy protections are implemented in practice. Ultimately, by facilitating interdisciplinary collaboration and ongoing research, we can ensure that LLMs are deployed in ways that prioritize privacy, transparency, and accountability.

6 Conclusion

This survey provided a comprehensive overview of the privacy risks associated with LLMs, focusing on privacy leakage and privacy attacks, as well as the defenses available to mitigate these risks. We systematically discussed the various ways in which LLMs can inadvertently expose sensitive information through mechanisms such as model inversion, training data extraction, and membership inference. Additionally, we categorized and reviewed existing privacy preservation techniques, including inference detection, federated learning, and confidential computing, evaluating their strengths and limitations. Another key contribution of this survey is the identification of practical challenges in implementing effective privacy protections. Furthermore, we outlined future research directions, emphasizing the need for more scalable, transparent, and efficient privacy solutions. By synthesizing current research, we aim to provide a clearer understanding of the privacy landscape in LLMs and guide future efforts to develop privacy-conscious AI systems.

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Data Availability The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare that they have no competing interests.

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